

	$\mathcal{K}_{0^+}^{\#1}$	$\mathcal{K}_{0^+}^{\#1}$	$\mathcal{K}_{0^+}^{\#1}$	$\mathcal{K}_{0^+}^{\#1}$
$\mathcal{K}_{0^+}^{\#1} \dagger$	0	0		
$\mathcal{K}_{0^+}^{\#1} \dagger^{\alpha\beta\chi}$	0	$3 k^2 \frac{(4)}{\kappa_{15}}$	$\mathcal{K}_{1^+}^{\#1} \alpha\beta$	$\mathcal{K}_{1^+}^{\#2} \alpha\beta$
$\mathcal{K}_{1^+}^{\#1} \dagger^{\alpha\beta}$	$-\frac{\gamma_1 k^2}{6}$	$\frac{\gamma_1 k^2}{3\sqrt{2}}$	0	0
$\mathcal{K}_{1^+}^{\#2} \dagger^{\alpha\beta}$	$\frac{\gamma_1 k^2}{3\sqrt{2}}$	$-\frac{\gamma_1 k^2}{3}$	0	0
$\mathcal{K}_{1^+}^{\#1} \dagger^\alpha$	0	0	$-k^2 \frac{(4)}{\kappa_2}$	$\frac{k^2 \frac{(4)}{\kappa_2}}{\sqrt{2}}$
$\mathcal{K}_{1^+}^{\#2} \dagger^\alpha$	0	0	$\frac{k^2 \frac{(4)}{\kappa_2}}{\sqrt{2}}$	$-\frac{k^2 \frac{(4)}{\kappa_2}}{2}$
			$\mathcal{K}_{2^+}^{\#1} \alpha\beta$	$\mathcal{K}_{2^+}^{\#1} \alpha\beta\chi$
			$\mathcal{K}_{2^+}^{\#1} \dagger^{\alpha\beta}$	$\frac{3\gamma_1 k^2}{2}$
			$\mathcal{K}_{2^+}^{\#1} \dagger^{\alpha\beta\chi}$	0

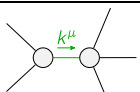
	$\mathcal{J}_{0^+}^{\#1}$	$\mathcal{J}_{0^+}^{\#1}$	$\mathcal{J}_{0^+}^{\#1}$	$\mathcal{J}_{0^+}^{\#1}$
$\mathcal{J}_{0^+}^{\#1} \dagger$	0	0		
$\mathcal{J}_{0^+}^{\#1} \dagger^{\alpha\beta\chi}$	0	$\frac{1}{3 k^2 \frac{(4)}{\kappa_{15}}}$	$\mathcal{J}_{1^+}^{\#1} \alpha\beta$	$\mathcal{J}_{1^+}^{\#2} \alpha\beta$
$\mathcal{J}_{1^+}^{\#1} \dagger^{\alpha\beta}$	$\frac{1}{3 \text{Det}(1^+)}$	$-\frac{\sqrt{2}}{3 \text{Det}(1^+)}$	0	0
$\mathcal{J}_{1^+}^{\#2} \dagger^{\alpha\beta}$	$-\frac{\sqrt{2}}{3 \text{Det}(1^+)}$	$\frac{2}{3 \text{Det}(1^+)}$	0	0
$\mathcal{J}_{1^+}^{\#1} \dagger^\alpha$	0	0	$-\frac{4}{9 k^2 \frac{(4)}{\kappa_2}}$	$\frac{2\sqrt{2}}{9 k^2 \frac{(4)}{\kappa_2}}$
$\mathcal{J}_{1^+}^{\#2} \dagger^\alpha$	0	0	$\frac{2\sqrt{2}}{9 k^2 \frac{(4)}{\kappa_2}}$	$-\frac{2}{9 k^2 \frac{(4)}{\kappa_2}}$
			$\mathcal{J}_{2^+}^{\#1} \alpha\beta$	$\mathcal{J}_{2^+}^{\#1} \alpha\beta\chi$
			$\mathcal{J}_{2^+}^{\#1} \dagger^{\alpha\beta}$	$\frac{1}{\text{Det}(2^+)}$
			$\mathcal{J}_{2^+}^{\#1} \dagger^{\alpha\beta\chi}$	0

Abbreviations used in matrices

$$\gamma_1 = \frac{(4)}{\kappa_1} + \frac{(4)}{\kappa_2} \quad \&\& \text{Det}(1^+) = -\frac{1}{2} k^2 \left(\frac{(4)}{\kappa_1} + \frac{(4)}{\kappa_2} \right) \quad \&\& \text{Det}(2^+) = \frac{3}{2} k^2 \left(\frac{(4)}{\kappa_1} + \frac{(4)}{\kappa_2} \right)$$

Lagrangian

$$\begin{aligned} & \frac{(4)}{\kappa_1} \partial_\beta \mathcal{K}_{\chi\delta}^\delta \partial^{\alpha\beta} \mathcal{K}_\alpha^\delta + \frac{(4)}{\kappa_2} \partial_\chi \mathcal{K}_{\beta\delta}^\delta \partial^{\alpha\beta} \mathcal{K}_\alpha^\delta - \frac{4}{3} \frac{(4)}{\kappa_1} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\alpha\chi}^\delta + 2 \frac{(4)}{\kappa_{15}} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\alpha\chi}^\delta - \\ & \frac{4}{3} \frac{(4)}{\kappa_2} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\alpha\chi}^\delta + \frac{2}{3} \frac{(4)}{\kappa_1} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}^\delta - 4 \frac{(4)}{\kappa_{15}} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}^\delta + \frac{2}{3} \frac{(4)}{\kappa_2} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}^\delta - \\ & \frac{5}{3} \frac{(4)}{\kappa_1} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\chi\alpha}^\delta - 2 \frac{(4)}{\kappa_{15}} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\chi\alpha}^\delta - \frac{5}{3} \frac{(4)}{\kappa_2} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\chi\alpha}^\delta - \frac{1}{3} \frac{(4)}{\kappa_1} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}^\delta - \\ & \frac{(4)}{\kappa_{15}} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}^\delta - \frac{1}{3} \frac{(4)}{\kappa_2} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}^\delta + \frac{(4)}{\kappa_{15}} \partial_\delta \mathcal{K}_{\alpha\beta\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi} - 2 \frac{(4)}{\kappa_{15}} \partial_\delta \mathcal{K}_{\beta\alpha\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi} \end{aligned}$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi} \mathcal{J}_{\alpha\beta\chi}$		
Source constraint(s)	# constraint(s)	Covariant form	
$\mathcal{J}_{0^+}^{\#1} = 0$	1	$\partial_\beta \mathcal{J}_\alpha^{\alpha\beta} = 0$	
$\mathcal{J}_{1^+}^{\#1\alpha} + 2 \mathcal{J}_{1^+}^{\#2\alpha} = 0$	3	$\partial_\chi \partial^\alpha \mathcal{J}_\beta^{\beta\chi} + \partial_\chi \partial^\chi \mathcal{J}_\beta^{\beta\alpha} = 3 \partial_\chi \partial_\beta \mathcal{J}^{\beta\alpha\chi}$	
$2 \mathcal{J}_{1^+}^{\#1\alpha\beta} + \mathcal{J}_{1^+}^{\#2\alpha\beta} = 0$	3	$\partial_\chi \mathcal{J}^{\alpha\beta\chi} + \partial_\chi \mathcal{J}^{\chi\alpha\beta} = \partial_\chi \mathcal{J}^{\beta\alpha\chi}$	
$\mathcal{J}_2^{\#1\alpha\beta\chi} = 0$	5	$3 \partial_\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\delta\beta\epsilon} + 3 \partial_\epsilon \partial^\epsilon \partial^\alpha \mathcal{J}^{\delta\beta}_\delta + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\chi\alpha\delta} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\chi\alpha\delta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\delta\alpha\chi} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\beta\alpha\delta} +$ $4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\delta\alpha\beta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\alpha\beta\chi} + 3 \eta^{\beta\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\alpha \mathcal{J}^{\delta\epsilon}_\delta + 3 \eta^{\alpha\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial_\delta \mathcal{J}^{\delta\beta\epsilon} + 3 \eta^{\beta\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\epsilon \mathcal{J}^{\delta\alpha}_\delta =$ $3 \partial_\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\delta\alpha\epsilon} + 3 \partial_\epsilon \partial^\epsilon \partial^\beta \mathcal{J}^{\delta\alpha}_\delta + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\beta\chi\delta} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\chi\beta\delta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\delta\beta\chi} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\alpha\beta\delta} +$ $2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\beta\alpha\chi} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\chi\alpha\beta} + 3 \eta^{\alpha\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\beta \mathcal{J}^{\delta\epsilon}_\delta + 3 \eta^{\beta\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial_\delta \mathcal{J}^{\delta\alpha\epsilon} + 3 \eta^{\alpha\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\epsilon \mathcal{J}^{\delta\beta}_\delta$	
Total # constraint(s):	12		
Resolved pole(s)	# polarization(s)	Square mass	Residue
	2	0	$\frac{1}{\frac{(4)}{\kappa_2}}$
Resolved unitarity condition(s):	$\frac{(4)}{\kappa_2} > 0$		